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MINI-PROJECT REPORT

Can PCA Reduce the Dimensionality of Mobile Feature Data While Preserving Important Information?

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Dataset: Kaggle Mobile Price Classification (train.csv, test.csv)
GitHub Repository: https://github.com/Krittish-Barman/College-Wca

Abstract

High-dimensional data is harder to store, visualize and model, so dimensionality reduction is a common step in machine learning. This project studies one question: can Principal Component Analysis (PCA) reduce the 20 technical features of mobile phones while preserving the information needed to identify a phone's price range? The Kaggle Mobile Price Classification dataset (2,000 labelled phones, 4 balanced price classes) was cleaned, standardized and explored using histograms, boxplots, a correlation heatmap and statistical measures (range, IQR, MAD, variance, standard deviation, Pearson correlation and outlier detection).

PCA was fitted on the standardized features, and the retained information was measured in two ways: explained variance and reconstruction error, and the predictive accuracy of a logistic regression classifier trained on the reduced data compared with the same classifier trained on all 20 features. The results show that PCA compresses this dataset only slightly. Sixteen components are needed to keep 90% of the variance and eighteen for 95%. With 17 components the classifier reaches 96.0% validation accuracy, close to the 96.25% obtained with all features, but with 10 components accuracy falls to 56% and with 2 components to 25.5%, which is no better than random guessing.

The cause is that most mobile features are almost uncorrelated, so there is little redundancy for PCA to remove, and the strongest price predictor, RAM, is independent of the other features and is pushed into a late component (PC6). It is concluded that PCA preserves the important information only when nearly all components are kept, and supervised methods such as feature selection or LDA are better suited to this problem.

Keywords: PCA, dimensionality reduction, mobile price classification, explained variance, logistic regression, scikit-learn

Introduction and Objective

Machine-learning datasets often contain many features, and some of them carry overlapping information. PCA is an unsupervised linear technique that rotates the data onto new, uncorrelated axes called principal components, ordered by the amount of variance each one explains. Keeping only the first k components gives a smaller dataset that retains as much variance as possible.

The main question of this project is: Can PCA reduce the dimensionality of mobile feature data while preserving important information? The work follows the complete machine-learning pipeline: data loading, preprocessing, exploratory data analysis, statistical analysis, model training, prediction, evaluation, visualization and interpretation. All code is available in the project repository: https://github.com/Krittish-Barman/College-Wca

Dataset Description

The dataset is the Mobile Price Classification dataset from Kaggle. The file train.csv has 2,000 rows and 21 columns (20 features and the target price_range), and test.csv has 1,000 rows with an id column but no target, so it can be used only to generate predictions. The target has four classes (0 = low, 1 = medium, 2 = high, 3 = very high cost) with exactly 500 phones in each, so the data is perfectly balanced and accuracy is a fair metric.

- **Numerical features (14):** battery_power, clock_speed, fc (front camera), int_memory, m_dep, mobile_wt, n_cores, pc (primary camera), px_height, px_width, ram, sc_h, sc_w, talk_time.
- **Binary features (6):** blue, dual_sim, four_g, three_g, touch_screen, wifi (already stored as 0/1).

Methodology

Data preprocessing

- **Missing values and duplicates:** none were found, so no rows were removed or imputed.
- **Encoding:** all columns are numeric and the six categorical flags are already 0/1, so no further encoding was required.
- **Inconsistent values:** px_height had 2 zeros and sc_w had 180 zeros. A phone cannot have a zero-pixel height or screen width, so these were treated as data errors and replaced with the median of the valid values. The median was used because it is not affected by extreme values, and deleting 9% of the rows would have lost information.
- **Train/validation split:** because test.csv has no labels, train.csv was split 80/20 with stratification to keep the classes balanced.
- **Scaling:** StandardScaler (mean 0, standard deviation 1) was fitted on the training part only, to avoid data leakage. Scaling is essential for PCA because it is based on variance, and without it RAM (variance about 1,176,000) would dominate features such as m_dep (variance about 0.08) only because of its units.

Exploratory data analysis and statistics

The target distribution, histograms, boxplots by price class, a correlation heatmap, scatter and pair plots and a feature dendrogram were used to understand the data. For the numerical features the range, interquartile range (IQR), mean absolute deviation (MAD), variance, standard deviation and the Pearson correlation coefficient (PCC) with price were calculated. Outliers were checked with the IQR rule and the Z-score rule.

Because PCA works by combining correlated features, the correlation structure is the most important result of this stage, and it is shown in Figure 1.

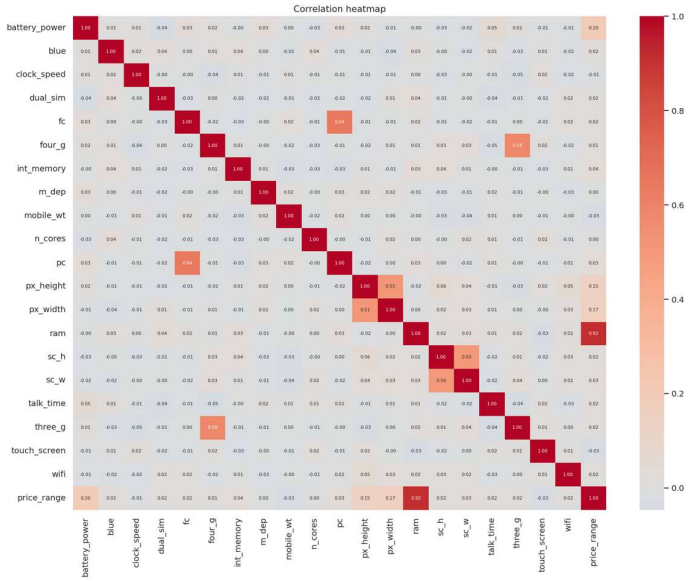


Figure 1: Correlation heatmap of the mobile features

Only four feature pairs are noticeably correlated: fc and pc (0.64), four_g and three_g (0.58), px_height and px_width (0.51), and sc_h and sc_w (0.51). All other correlations are close to zero. RAM has a correlation of 0.917 with price, which is far stronger than any other feature. Table 1 gives the descriptive statistics for selected features.

Table 1: Statistical measures of selected numerical features

Feature	Range	IQR	MAD	Variance	Std	PCC (price)
ram	3742	1857.0	937.0	1,176,644	1084.7	0.917
battery_power	1497	763.5	382.0	193,088	439.4	0.201
px_width	1498	758.3	372.9	186,796	432.2	0.166
px_height	1959	663.3	365.9	196,531	443.3	0.150
m_dep	0.9	0.6	0.25	0.083	0.288	0.001

Outliers were rare: the IQR rule flagged 18 values in fc and 2 in px_height, and the Z-score rule flagged 12 values in fc. These are real phones (for example, high-resolution cameras), not errors, so they were kept.

Principal Component Analysis (model)

PCA computes the eigenvectors of the covariance matrix of the standardised data. Each eigenvector is a principal component, and its eigenvalue is the variance it explains. The number of components to keep was judged using the scree plot, the cumulative explained variance (80%, 90%, 95% thresholds) and the Kaiser rule (eigenvalue greater than 1). The loadings were examined to see which original features form each component.

Evaluation strategy

To test whether important information is preserved, two checks were used. First, the reconstruction error (mean squared error between the original data and the data rebuilt from k components) measures the information lost. Second, a logistic regression classifier was trained on the original 20 features and on PCA features with different numbers of components, and compared using accuracy, precision, recall, F1-score (macro-averaged), ROC-AUC and the confusion matrix. Logistic regression was chosen because it is simple and stable, so any change in performance comes from the features and not from model complexity. Finally, GridSearchCV with 5-fold cross-validation tuned the number of PCA components (2 to 20) and the regularisation strength C.

Results and Analysis

Explained variance

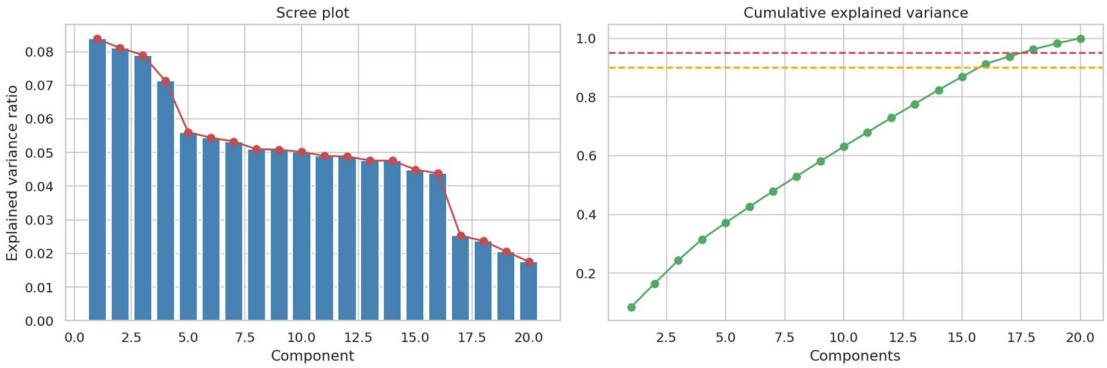


Figure 2: Scree plot and cumulative explained variance

The first component explains only about 8% of the variance and the scree plot has no sharp elbow, so the variance is spread almost evenly over the components. Fourteen components are needed to keep 80% of the variance, 16 for 90% and 18 for 95%. Ten components have an eigenvalue greater than 1 (Kaiser rule). This confirms the expectation from the correlation heatmap: there is little redundancy for PCA to remove.

Loadings and 2-D projection

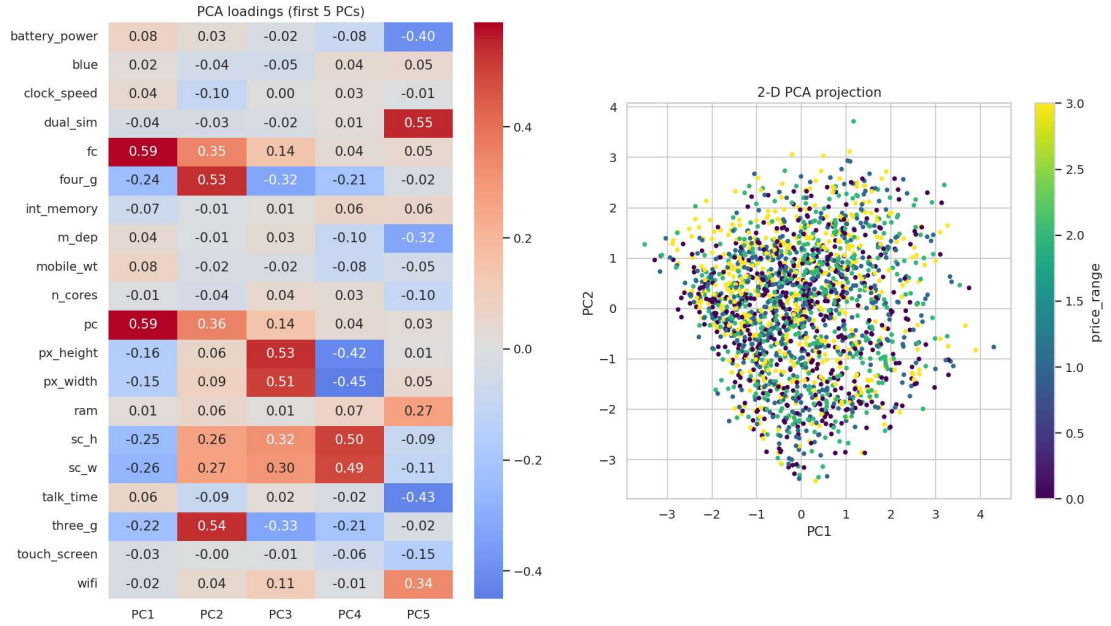


Figure 3: PCA loadings of the first five components (left) and 2-D projection coloured by price class (right)

Each early component mainly captures one correlated pair: PC1 is the cameras (fc, pc), PC2 is the networks (3G, 4G), PC3 is pixel resolution and PC4 is screen size. RAM hardly appears in these components and has its largest loading on PC6. In the 2-D projection the four price classes are completely mixed, so the first two components do not keep the information that separates the price classes.

Information preserved: model comparison

Table 2: Logistic regression on all features and on PCA features (validation set)

Features used	Variance kept	Accuracy	Precision	Recall	F1	ROC-AUC
All 20 features	100%	0.9625	0.9626	0.9625	0.9625	0.9986
PCA, 17 comps	94%	0.9600	0.9602	0.9600	0.9598	0.9986
PCA, 15 comps	87%	0.7950	0.7983	0.7950	0.7964	0.9509
PCA, 10 comps	63%	0.5600	0.5485	0.5600	0.5527	0.8221
PCA, 5 comps	37%	0.3275	0.3202	0.3275	0.3197	0.5772
PCA, 2 comps	16%	0.2550	0.1869	0.2550	0.2108	0.5430

Accuracy collapses when only a few components are kept: with 2 components it is 25.5%, equal to random guessing among four classes. It recovers only when about 17 components are used, where it matches the baseline. Hyperparameter tuning selected 17 components and C = 10, giving a cross-validated accuracy of 95.9% and a validation accuracy of 97.25%. The confusion matrices of the baseline and the tuned PCA model are strongly diagonal, and the few mistakes occur between neighbouring price classes.

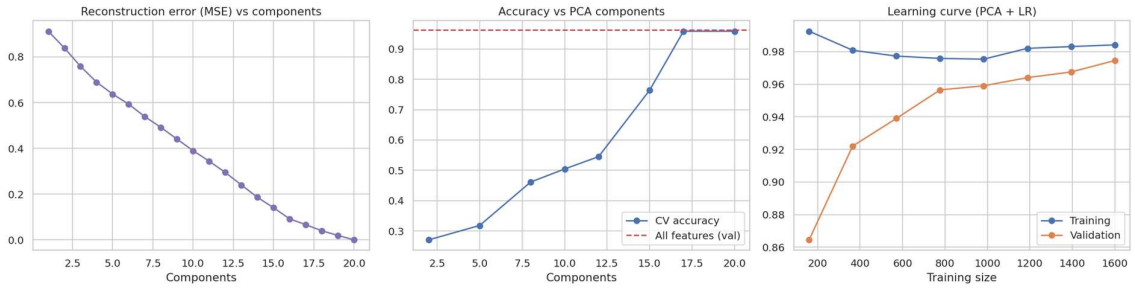


Figure 4: Reconstruction error vs components, accuracy vs components, and learning curve

The reconstruction error falls steadily and almost linearly as components are added, again showing that no small group of components holds most of the information. The accuracy curve rises sharply between 10 and 17 components. In the learning curve the training and validation accuracies are close and high, so the model is not over-fitted. A K-Means clustering of the PCA space gave a silhouette score of only about 0.09, which means the price classes do not form natural clusters without the labels.

Interpretation and Discussion

- **What did the model learn?** The mobile features are mostly independent of each other, and price depends mainly on RAM, followed by battery power and screen resolution.
- **Most important features:** RAM (PCC 0.917), then battery_power, px_width and px_height. In the PCA structure the main groups are cameras, 3G/4G, pixel resolution and screen size.
- **Why does PCA fail to reduce much?** PCA ranks components by variance, not by usefulness for the target. RAM is the best price predictor but is uncorrelated with the other features, so it falls into PC6, which low-dimensional versions discard.
- **Limitations:** PCA is unsupervised and linear, its components are harder to interpret than the original features, the dataset is clean and looks synthetic, and test.csv has no labels, so evaluation used a split of train.csv.
- **Possible improvements:** use supervised reduction (LDA or feature selection on RAM, battery power and pixel size), apply PCA only to the correlated groups, try non-linear classifiers, or use t-SNE/UMAP for visualisation.
- **Real-world use:** a manufacturer or retailer could estimate the price category of a new phone from its specifications, or check whether a phone is priced too high for its features.

Conclusion

This project asked whether PCA can reduce the dimensionality of mobile feature data while preserving important information.

The answer is: Only slightly. Sixteen to eighteen of the twenty components are needed to keep 90–95% of the variance, and about 17 are needed to match the accuracy of the full-feature model (96.0% against 96.25%; 97.25% after tuning). Aggressive reduction loses the important information, with accuracy of 56% at 10 components and 25.5% at 2 components.

The reason is the structure of the data. Most features are nearly uncorrelated, so PCA has little redundancy to remove, and the key feature RAM is independent of the others. PCA is therefore useful here only for merging the four correlated pairs (camera, network, resolution and screen size), and for this prediction task, feature selection or supervised dimensionality reduction would be a better choice.

Project Repository

Source code, notebook and dataset: https://github.com/Krittish-Barman/College-Wca

To reproduce the results, open the notebook in Google Colab, upload train.csv and test.csv when asked, and run all cells in order. The random seed is fixed at 42.

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